GENERAL THETA FOUNDATIONS I: ACQUIRED GEOMETRY AND CAUSAL RESOURCE TRANSFER

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Abstract. For known prepared positive instruments with a finite separating affine realization of executable tests, we prove matching finite-state and checkpoint excess-risk bounds under jointly optimized acquisition and stopping. The assumptions concern affine unnormalized prediction, actual one-call mass, a global cover and either an interior buffer or a binary boundary certificate; no separate mixing certificate or externally imposed stopping hazard is required. With positive acquisition cost, both excess risks are comparable to $M^{-2/d}$. The upper is implemented by a homogeneous machine with all preparation, continuing and halting labels counted. Its proof calibrates a killed first-run occupation and absorbs its curvature error using the acquisition cost. The converse bounds the stopped law by actual occupation mass, uniformly over adaptive stopping rules. Gaussian hidden-state observations and noncommuting quantum instruments verify the same theorem. For independent prepared observations we also characterize exactly every obtainable terminal law under a mean acquisition budget, derive a joint acquisition–memory curve and exact small-state optima, and give an attained calibration–deficiency extension. Informative boundary-reaching updates without a strict uniform same-report contraction, singular prepared sectors, cost-aware causal simulation and finite-bit synthesis under effective primitive certificates are treated explicitly.

1. Introduction

The resource content of a causal experiment is not determined by its formal parameter space. Preparation and intervention determine which histories can occur; executable future tests identify the predictions carried by those histories; and an observer must update its finite memory using only the reports it has actually seen. If acquisition may stop on the basis of those reports, a further difficulty appears. The terminal predictive law is selected by the stopping rule and cannot be replaced by either a stationary distribution or an unconditional one-step law.

We study this problem for known positive instruments admitting a finite separating affine realization of their future tests. Our principal result optimizes both sensing and stopping. The sampling cost is strictly positive, but the number of calls has no deterministic cap and no externally imposed hazard. The finite observer is autonomous: its complete state contains every continuing and halting label, and its decoder receives no uncharged time index or current report. Under explicit actual-row mass and covering assumptions, the best such observer and the checkpoint both lose order $M^{-2/d}$ relative to the optimal full-history stopping experiment. The exponent is established from the acquired laws of executable tests, not from a supplied covariance or a parameter manifold.

Two obstacles distinguish this statement from a static quantization bound. First, quantizing the belief at each time does not automatically produce compatible conditional means for one reused finite-state machine. Second, the expected number of visits may grow when the quantizer or stopping policy changes. Our construction introduces a small private safety probability, solely as part of the candidate machine. A finite-dimensional fixed point calibrates all of its conditional means to a preparation-injected first-run occupation. A resolvent identifies that occupation with the actual execution. The Bellman error is then a single mass-weighted Jensen loss. Positive

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acquisition cost absorbs this loss and bounds the expected number of visits independently of the safety probability. The benchmark remains unrestricted by that auxiliary probability.

The converse uses the same resource. For every adaptive stopping experiment, its terminal predictive law is dominated by the one-call density bound multiplied by its actual expected number of calls. This factor is indispensable. Near optimality and positive sampling cost bound it, and a small-ball argument applies to every finite decoder. Thus recursive implementation and the decision-theoretic lower bound meet at occupation mass, rather than at a formal reachable set.

Theorem 3.1 is the general result. Theorems 5.1 and 5.3 verify it directly from two different primitive instruments. A binary Gaussian example has genuine, positive-probability continuation and stopping regions. Theorem 6.1 reaches both boundary states and admits informative, locally expanding updates without a strict uniform same-report contraction. Theorem 6.2 treats immutable prepared strata of different dimensions without a global mixing condition. Theorem 7.1 gives an exact necessary-and-sufficient acquired-law criterion for independent prepared observations, including arbitrary singular preparations. It yields an explicit joint mean-acquisition–memory law, not merely a Lagrange-multiplier interpretation of the first theorem. Corollary 8.1 adds attained calibration and simulation defects in that same task.

Belief-state reduction, conditional squared-loss projection, optimal stopping and occupation measures are classical tools. Value-directed compression and quantized partially observed stopping have substantial prior literature [5, 7, 8, 3]. The preparation-injected calibration and boundary-valid mass–curvature estimate used below also have explicit predecessors in the retained technical articles [6]. The new combination is the cost-coercive, endogenous-stopping construction and its all-stopping converse, together with an exact acquired-law/total-state frontier for the independent prepared class. Section 9 separates these claims from standard tools and from questions not settled here.

2. Prepared experiments and executable prediction

2.1. **Primitive instruments.** All report and hidden spaces are standard Borel. A preparation fixes a known probability law of the hidden system. Actions form a finite set A, available after every live history. A raw action a is a normalized nonnegative kernel for the next hidden state and its report. The current report, fresh private coins and all previously retained labels are available to the observer; future reports and the protected terminal audit are not. There is no unknown parameter supplied to the program. Uniform simulation across an unknown parameter family is a separate, explicitly typed comparison in Section 8.

We impose the following finite realization assumption on the primitive experiment. A compact convex base $B \subset \mathbb{R}^d$ and its cone

$$C_B = \{w(1,u) : w \geq 0,\ u \in B\}$$

realize a finite family of bounded executable future tests, together with the constant test. Their linear span is closed under every report-event instrument and separates the predictive states. On this cone the raw instrument has an integrable positive linear density $L_a(y)$ with respect to a fixed sigma-finite reference λ_a:

$$L_a(y)(1,u) = (\ell_a(y,u), U_a(y,u)), \qquad \int \ell_a(y,u)\, d\lambda_a(y) = 1. \tag{1}$$

Integrability means $\int \|L_a(y)\|\, d\lambda_a(y) < \infty$ in a fixed finite-dimensional operator norm. The normalized update and its actual one-call law are

$$F_a(y,u) = \frac{U_a(y,u)}{\ell_a(y,u)}, \qquad \mu_{a,u} = F_a(\cdot,u)_{\#}(\ell_a(\cdot,u)\lambda_a), \qquad K_a f(u) = \int f\, d\mu_{a,u}. \tag{2}$$

Only positive-likelihood points are used. Common null versions are fixed uniformly on the full compact domain. At zero likelihood, positivity of the cone and bounded tests force the numerator to vanish; the quantitative main theorem below assumes positive likelihood on its operating domain.

The realization is a hypothesis to be verified from the raw kernel, not a conclusion from physical finite dimension alone. For the standard Borel quotient assertion we additionally require the attained evaluation image to be Borel and to admit a Borel history section. The estimates below do not give the observer access to that section. The mandatory-first-call status is part of the visible interface and is not identified with the post-acquisition status.

Lemma 2.1 (Predictive quotient and conditional integration)**.** *Within each fixed legal-interface status, the conditional executable-test vector X_t realizes future-test equivalence measurably. Two such reachable histories have the same vector exactly when all their legal bounded future tests agree. The full predictive object also records the initial mandatory-call, continuing or halting interface status. Under the declared Borel-image and section condition it is minimal among standard Borel predictive realizations with the same condition. If S is any finite retained variable, including allowed past private coins, and $v_s = \mathbb{E}[X \mid S = s]$, then*

$$\mathrm{Law}(Y \mid S = s, A = a) = \ell_a(y, v_s)\lambda_a(dy), \qquad \mathbb{E}[X' \mid S = s, A = a, Y] = F_a(Y, v_s). \tag{3}$$

The same assertions hold for finite unnormalized occupation ensembles.

Proof. The conditional expectations of the finitely many bounded tests are Borel versions, and their recursive joint numerator and denominator are (1). Equality of these coordinates gives equality of all finite executable continuations by event pullback and induction. The declared bounded monotone closure gives the other allowed tests. The separating tests prove the reverse implication. For a competing realization $T : H \to Z$ with Borel attained image Z_0 and Borel section $\sigma : Z_0 \to H$, the factor is $X \circ \sigma$ on Z_0, and equality of predictive fibers gives $X = (X \circ \sigma) \circ T$. This proves the asserted minimality. We do not infer a Borel image or section for an arbitrary analytic quotient.

For (3), integrate the affine numerator and denominator over the histories in the positive-mass label. This gives $\ell_a(y, v_s)$ and $U_a(y, v_s)$ before division. Division under the new report law proves the identity. Independent action coins chosen from s do not reveal hidden history. Homogeneity gives the unnormalized version. This is conditional integration over report events, not conditioning a continuous report on a positive-probability singleton. In general $\mathbb{E}[F_a(y, X) \mid S]$ under the old law is not $F_a(y, v_s)$. □

2.2. **Geometry and a protected task.**

Definition 2.2 (Regular actual-row geometry)**.** There are full-dimensional compact convex sets $K \subset D \subset B$. One of two geometric certificates is required: either an interior buffer $K + \overline{B}_r \subset D$ for some $r > 0$, or the binary boundary certificate $d = 1$, $K = D = B = [0, 1]$ with a Bernoulli squared audit $g(u) = u(1 - u)$. The latter replaces, rather than asserts, an interior buffer. The known preparation u_0 is in D. Every normalized raw output from D lies in K. Off a common reference-null set the likelihood is positive on D. For every action and every $u \in D$,

$$\mu_{a,u} \le \Lambda \,\mathrm{vol}_d\,|_K, \qquad \mathrm{TV}(\ell_a(\cdot, u)\lambda_a, \ell_a(\cdot, v)\lambda_a) \le L_{\mathrm{rep}}\|u - v\|. \tag{4}$$

The rows are weakly continuous. A fixed bounding-box grid, intersected with K, supplies at most m cells of diameter $C_K m^{-1/d}$ with convex closures and row-null boundaries. All constants and the domain are fixed before selecting the observer or its budget.

The grid is global on the declared operating domain, but all mass estimates use actual rows. The dimension in (4) is that of separating executable predictions whose acquired law has the displayed density. Positivity alone does not imply this property. No separate mixing, feedback-uniform overlap or summable posterior-derivative certificate is used. The buffer is a substantive condition and can itself imply cone-metric contraction; it must not be described as a class with no stability implications. The alternative binary boundary certificate does not impose that buffer and has a nonuniformly contracting, locally expanding realization in Theorem 6.1.

After commitment the experiment performs one fixed protected bounded audit Y. In the chosen task coordinates,

$$\mathbb{E}[Y \mid X = u] = u, \quad \mathbb{E}[\|Y\|^2 \mid X = u] = c_0(u), \quad g(u) = c_0(u) - \|u\|^2, \tag{5}$$

where c_0 is affine. Admissible readouts lie in a fixed compact convex set containing B. We normalize a bound $0 \le \|Y - r'\|^2 \le L_0$ on that set. Thus $0 \le g \le L_0$, and the full-history Bayes loss is g. A fixed positive definite task metric is reduced to this notation by an invertible linear change of coordinates. A finite randomized audit of noncommuting coordinates is allowed; simultaneous measurement of those coordinates is not required.

2.3. **Observer and resource conventions.** At least one raw acquisition is mandatory. Every acquisition costs $\kappa > 0$. After a report the observer may stop or continue; it must stop almost surely and have finite mean call count $\mathbb{E}\tau$. The objective is

$$J_\kappa = \kappa \mathbb{E}\tau + \mathbb{E}\|Y_\tau - r_{S_\tau}\|^2. \tag{6}$$

A full-history strategy may choose each sensing action and its stopping decision from its complete observed history and fresh coins. A homogeneous autonomous observer has at most M labels, with a fixed continuing/halting status for each label, a fixed action/update rule on continuing labels and a fixed readout on halting labels. No time argument, previous report, sector mark or private random seed is retained outside those labels. A report may be routed directly to a halting label. A temporary calculation and fresh coins at a live decision are allowed, but their workspace is erased and separately charged. A preparation label counts even when used once.

A checkpoint has the full stopped history once and compresses it into at most M terminal labels. Its previous full-history storage is not asserted to fit in M. It uses the same acquisition and stopping class as the full-history benchmark. Preparation and the final protected audit each count as one additional physical call when total physical acquisitions are reported. They are excluded from τ and from the varying part of (6); adding their common fixed cost changes none of the excess risks.

3. The cost-coercive acquired-geometry theorem

For a probability ν in the task space put

$$q_m(\nu) = \inf_{r_1,\dots,r_m} \int \min_j \|x - r_j\|^2 \, d\nu(x).$$

Let $V_\kappa(u_0)$ be the full-history infimum of (6). Write $\mathcal{R}_{\mathrm{cp}}$ and $\mathcal{R}_{\mathrm{aut}}$ for the corresponding checkpoint and autonomous infima.

Theorem 3.1 (Joint acquisition and stopping with finite memory)**.** *Assume the raw realization, Definition 2.2 and* (5)*. For every fixed $\kappa > 0$ there are $0 < c_\kappa \le C_\kappa < \infty$, depending only on the declared experiment, task and κ, such that for every $u_0 \in D$ and $M \ge 3$,*

$$c_\kappa M^{-2/d} \le \mathcal{R}_{\mathrm{cp}}(\kappa, M; u_0) - V_\kappa(u_0) \le C_\kappa M^{-2/d}, \tag{7}$$

$$c_\kappa M^{-2/d} \le \mathcal{R}_{\mathrm{aut}}(\kappa, M; u_0) - V_\kappa(u_0) \le C_\kappa M^{-2/d}. \tag{8}$$

The upper in (8) *uses at most $1 + 2m$ labels, $m = \lfloor (M-1)/2 \rfloor$: one initial label, m live labels and m halting labels. It obeys $\mathbb{E}\tau \le 2(1 + L_0/\kappa)$, or stops after its first call in the coarse-budget case. Its sensing and stopping decisions are functions of those labels and allowed fresh coins. The benchmark optimizes all legal full-history sensing and stopping strategies; no prescribed hazard or fixed controller is imposed on it.*

The post-observation value is the bounded concave solution selected by finite-horizon approximation of

$$W_\kappa(u) = \min\{g(u),\ \kappa + \min_a K_a W_\kappa(u)\}, \qquad V_\kappa(u_0) = \kappa + \min_a K_a W_\kappa(u_0). \tag{9}$$

The exact checkpoint formula is

$$\mathcal{R}_{\mathrm{cp}}(\kappa, M; u_0) = \inf_{\pi,\tau}\{\kappa\mathbb{E}_\pi\tau + \mathbb{E}_\pi g(X_\tau) + q_M(\mathrm{Law}_\pi(X_\tau))\}. \tag{10}$$

In particular the geometric lower concerns the law actually selected by the acquisition and stopping rule.

The constants need not remain bounded as $\kappa \downarrow 0$. A positive price controls expected acquisition, not a deterministic maximum. Equation (8) alone does not imply equality between a constrained frontier and its Lagrange dual. Section 7 proves such a frontier directly for a specified nontrivial class, rather than inferring it from (9).

Lemma 3.2 (Bellman value and an internal safety regularization)**.** *The value in* (9) *is attained by a proper full-history Markov stopping policy. It satisfies* $0 \le W_\kappa \le L_0$ *and* $\mathrm{Lip}_K W_\kappa \le L_0/r$ *in the buffered case and* $\mathrm{Lip}_{[0,1]} W_\kappa \le 1$ *in the binary boundary case. For* $0 < s < 1$*, let* $\beta = 1 - s$ *and define*

$$W_s(u) = \min\{g(u),\ sg(u) + \beta[\kappa + \min_a K_a W_s(u)]\}, \qquad V_s = \kappa + \min_a K_a W_s(u_0). \tag{11}$$

Then W_s *is the unique bounded fixed point, is concave, has the same bounds and Lipschitz constant, and*

$$0 \le V_s - V_\kappa \le sL_0^2/\kappa. \tag{12}$$

Here s *is a private coin used only if the candidate policy elects to continue: with probability* s *it instead stops immediately. It is not an environmental stopping rule.*

Proof. Finite-horizon stopping values start with g and decrease under the operator in (9). They remain between zero and L_0. For concave f, its perspective $wf(v/w)$ is concave, so (1) makes $K_a f$ concave. The minimum of finitely many concave functions is concave. Monotone convergence passes the bounded limit through K_a and the finite minimum. In the buffered case, concavity on D bounds all slopes on K by L_0/r. In the binary case, $0 \le W_\kappa \le u(1-u)$ forces $W_\kappa(0) = W_\kappa(1) = 0$. Concavity bounds every secant slope between the endpoint one-sided slopes; $W_\kappa(u)/u \le 1 - u$ and $W_\kappa(u)/(1-u) \le u$ bound those slopes by 1 and -1. Thus W_κ is globally 1-Lipschitz on $[0,1]$. The limit is continuous on K in both cases. A minimizing finite-action selector is Borel. Along its continuation region, the Bellman equality gives $\kappa\mathbb{E}(\sigma \wedge t) \le W_\kappa(u) \le L_0$, where σ is the number of additional calls. Consequently $\mathbb{E}\sigma \le L_0/\kappa$. Stopping is proper, and the residual bounded value on $\{\sigma > t\}$ tends to zero. The equality identifies the value; the corresponding Bellman inequality bounds every legal history-dependent policy below it. Including the mandatory first call gives $\mathbb{E}\tau \le 1 + L_0/\kappa$ and $V_\kappa \le \kappa + L_0$.

The regularized operator is a sup-norm contraction of coefficient β. The same perspective argument gives concavity, and stopping immediately bounds its value by g. To compare it with the unregularized optimum, follow that optimum but add the independent safety coin at each of its intended continuation decisions. A first safety stop occurs with probability at most $s\mathbb{E}(\tau - 1) \le sL_0/\kappa$. On this event the increase in audit loss is at most L_0 and the acquisition cost decreases. This proves (12). The regularized class is a subclass of legal stopping strategies, proving the reverse inequality. No exchange of an infimum with a limit of unbounded stopping losses was used. □

4. Actual occupation, recursive calibration and the converse

Lemma 4.1 (Mass-weighted curvature estimate)**.** *Let* f *be concave and* L_f*-Lipschitz on the compact convex set* K*. For the common grid* $\mathcal{P}_m$ *and any finite measure* $\mu \le A\,\mathrm{vol}_d|_K$*,*

$$0 \le \sum_C \mu(C) f(\bar{x}_C) - \int f\,d\mu \le C_{K,d} A L_f m^{-2/d}. \tag{13}$$

Zero-mass cells contribute zero. Neither a smooth Bellman surface nor an interior cutoff of the actual measure is required.

Proof. For completeness we reproduce the boundary-valid argument used in the retained article [6]. Extend the convex function $-f$ to $\mathbb{R}^d$ by $\inf_{z\in K}\{-f(z)+L_f\|x-z\|\}$. This is the infimal convolution of the convex extended-value function $-f+\iota_K$ with $L_f\|\cdot\|$, hence is convex and globally Lipschitz. The Lipschitz inequality on K shows it agrees with $-f$ there, including boundary points. Its distributional Laplacian ξ is a positive measure. After mollification, subtract a supporting plane at the center z of a grid cube of side h. The resulting convex function w is nonnegative and vanishes at z. Convexity gives $w(z+R\theta)\le R\partial_R w(z+R\theta)$. A submean estimate and integration on a sphere with radius comparable to h therefore give

$$\sup_C w \le C_d h^{2-d}\xi(\overline{B}_{3\sqrt{d}h}(z)).$$

One may first average the sphere radius in $[2\sqrt{d}h, 3\sqrt{d}h]$; this avoids exceptional radii. In dimension one use the endpoint slope difference. The affine term cancels at the centroid, and $w(\bar{x}_C)\ge 0$. Multiplication by $\mu(C)\le Ah^d$ bounds the cell loss by $C_dAh^2\xi(\overline{B}_{3\sqrt{d}h}(z))$. The enlarged balls have bounded overlap. On a fixed neighborhood of K, a smooth cutoff χ gives $\int\chi\,d\xi\le L_f\int\|\nabla\chi\|$. Sum and pass through mollification. A bounding-box grid with $\lfloor m^{1/d}\rfloor$ subdivisions in each direction has at most m cells and $h\le C_K m^{-1/d}$. The finitely many small cases adjust the constant. □

Lemma 4.2 (Killed first-run calibration)**.** *Fix $s\in(0,1)$ and a common grid with m cells. There is a homogeneous observer with at most $1+2m$ labels and actual expected call count T whose post-report visit occupation has conditional representatives $q_i\in\overline{C}_i$. At q_i its stopping/action probabilities minimize* (11)*. If $R_s^{(m)}$ denotes its costed risk, then*

$$R_s^{(m)}-V_s=\sum_i w_iW_s(q_i)-\int W_s\,d\mu,\qquad \mu\le\Lambda T\,\mathrm{vol}_d\,|_K,\quad T=\sum_i w_i=\mathbb{E}\tau. \tag{14}$$

The measure μ is the actual unnormalized pooled-row prediction measure before cell routing. Its cell masses and barycenters are w_i, q_i. It is not asserted to be the full-history terminal predictive law.

Proof. Choose an action minimizing the first-call value V_s. Let b_i be its unnormalized predictive vector after routing the first report to the cell of $F_a(y,u_0)$; thus $\sum_i e(b_i)=1$, where e denotes cone mass. Consider the compact convex product set

$$\mathcal{Z}_s=\{(z_i): z_i\in C_{\overline{C}_i},\ \sum_i e(z_i)\le 1/s\}.$$

At each positive row write $z_i=w_i(1,q_i)$ and choose probabilities $p_{i0},(p_{ia})_{a\in A}$ supported on minimizers of the STOP and continuation alternatives in (11). At zero mass choose any representative in the cell; its contribution vanishes. The correspondence sends z to

$$z_j'=b_j+\beta\sum_{i,a}p_{ia}\int\mathbf{1}_{\{F_a(y,q_i)\in C_j\}}L_a(y)z_i\,d\lambda_a(y). \tag{15}$$

It maps into $\mathcal{Z}_s$ because $1+\beta/s=1/s$. Its values are nonempty, compact and convex: the minimizing choices are finite simplices and the displayed output is linear in their probabilities. Its graph is closed. At positive limiting mass, representatives converge and the continuous Bellman alternatives preserve minimizing support. Affine raw maps are continuous in u off common null sets. Positive likelihood and row-null cell boundaries permit dominated convergence of the cell indicators. At zero limiting mass the integrable operator bound kills the contribution uniformly. Kakutani's theorem [4] gives a fixed point.

Freeze its representatives and probabilities. Let P be the continuing ensemble map. Then $z = b + Pz$. Although normalization appears in selecting the table, P is now a positive linear map on the physical ensembles, with mass at most β on every positive input. Thus

$$z = (I - P)^{-1}b = \sum_{t\geq 0} P^t b. \tag{16}$$

This is exactly the first-run post-report occupation from the given preparation. It is not a stationary law inserted at initialization. The machine starts in one exclusive label; each acquired report enters a live cell label. At that label it stops with probability $h_i = p_{i0} + s\sum_a p_{ia}$, moving to its corresponding halting label, or acquires once with probability βp_{ia} using action a. The safety coin ensures proper stopping. The terminal vector is $h_i z_i$, so its centroid is also q_i. This equality holds because this program stops from the retained label; it is not assumed for arbitrary report-dependent stopping.

Let μ_0 be the first acquired law and set

$$\mu = \mu_0 + \beta \sum_{i,a} w_i p_{ia} \mu_{a,q_i}.$$

Equation (15) says precisely that its cell vectors are z_i. Conditional integration, Lemma 2.1, identifies it as the actual pooled-row law under visit occupation. Its mass is T and its density bound follows from (4). The Bellman equalities at q_i are

$$W_s(q_i) = h_i g(q_i) + \beta \sum_a p_{ia}(\kappa + K_a W_s(q_i)).$$

Multiply by w_i, sum, and use $\beta\sum_{i,a} w_i p_{ia} = T - 1$. The terminal readouts are calibrated, so the physical risk is $\kappa T + \sum_i h_i w_i g(q_i)$. Since $V_s = \kappa + \int W_s \, d\mu_0$, cancellation gives (14). This is one aggregate identity over actual visits; no fixed-time calibration is asserted. □

Upper bound in Theorem 3.1. Lemmas 3.2–4.2 give, with a constant G independent of s, m, κ, u_0,

$$\kappa T \leq R_s^{(m)} \leq V_s + GTm^{-2/d}, \qquad V_s \leq \kappa + L_0. \tag{17}$$

If $Gm^{-2/d} \leq \kappa/2$, the acquisition cost absorbs the occupation error:

$$T \leq 2(1 + L_0/\kappa), \qquad R_s^{(m)} - V_\kappa \leq sL_0^2/\kappa + 2G(1 + L_0/\kappa)m^{-2/d}.$$

Choose any dyadic $0 < s \leq \min(1/2, m^{-2/d})$. Its program description uses $O(\log m)$ bits; it uses no clock states. If $Gm^{-2/d} > \kappa/2$, simply acquire once and stop. The excess is at most L_0, which is at most $(2GL_0/\kappa)m^{-2/d}$. The claimed state count and expected-call bound follow, and $m \geq M/4$ for $M \geq 3$ converts the rate.

For the checkpoint use an optimal full-history policy from Lemma 3.2, followed by a global grid. The full-history projection gives (10), and the grid adds at most $C_K^2 M^{-2/d}$. Randomized encoders cannot beat the pointwise nearest-center loss. Conversely optimal codebooks exist on the compact task range, so the formula is exact. □

Lemma 4.3 (Stopped actual-mass converse)**.** *For every legal adaptive sensing/stopping policy with $T = \mathbb{E}\tau < \infty$, including all observer coins, its full-history terminal law satisfies*

$$\nu(B') = \mathbb{P}(X_\tau \in B') \leq \Lambda T \operatorname{vol}_d(B' \cap K). \tag{18}$$

If a terminal decoder has at most M values, its costed excess $e = J_\kappa - V_\kappa$ obeys

$$e \geq a_d(\Lambda M T)^{-2/d}, \qquad e(V_\kappa + e)^{2/d} \geq a_d(\kappa/(\Lambda M))^{2/d}, \tag{19}$$

where $a_d = \frac{1}{2}(2v_d)^{-2/d}$ and v_d is unit-ball volume.

Proof. The event of making call t is measurable before its report. Ignore the later stop selection, bound the actual row at that history by (4), and sum:

$$\nu(B') \le \sum_{t\ge 1} \mathbb{E}[\mathbf{1}_{\{\tau\ge t\}}\mu_{A_{t-1},X_{t-1}}(B')] \le \Lambda T \operatorname{vol}_d(B'\cap K).$$

Tonelli justifies the sum. This does not bound the law conditioned on stopping by an unconditional probability row. At most $\Lambda T M v_d h^d$ of ν lies within distance h of any M codepoints. Choosing that expression to equal $1/2$ gives the first inequality. Orthogonal projection on the full stopped history shows that quantization is added to the costed full-history risk, which is at least V_κ. Finally $\kappa T \le J_\kappa = V_\kappa + e$ gives the second inequality. The proof applies to history-dependent stopping, randomized finite observers and checkpoint encoders alike. □

Completion of Theorem 3.1. If $e \le L_0$, then $T \le 1 + 2L_0/\kappa$, since $V_\kappa \le \kappa + L_0$. Apply Lemma 4.3. If $e > L_0$, shrink the lower constant and use $M^{-2/d} \le 1$. This gives a positive uniform c_κ for all policies and hence both lower bounds. The upper constructions were already proved. The theorem uses the stability supplied by coercive cost and actual visit mass, not a hypothesis that the predictive update contracts. □

5. Two raw-kernel realizations of the same theorem

5.1. **Controlled Gaussian hidden-state observations.** Let the hidden state take values in $\{1,\dots,q\}$, $q \ge 2$. Fix a positive invertible column-stochastic matrix P with $P_{ij} \ge a_0 > 0$. A sensing action a selects the report law

$$Y \mid Z = i \sim N(m_i^a, \Sigma_a), \qquad Y \in \mathbb{R}^{q-1},$$

where Σ_a is positive definite and the $q-1$ vectors $m_i^a - m_q^a$ are independent. After the report the hidden state moves by P. All data are known; the finite sensor library is optimized by the observer. The protected audit is the new hidden-state category under squared probability prediction.

Theorem 5.1 (Gaussian optimal acquisition and stopping)**.** *This primitive experiment satisfies Theorem 3.1 with $d = q-1$, for every fixed $\kappa > 0$, every initial belief in $D = \{u \in \Delta_q : u_i \ge a_0/2\}$, and all $M \ge 3$. No report is truncated, no Gaussian tail is discarded, and the comparator jointly optimizes sensing and stopping.*

Proof. In probability coordinates the positive raw instrument is $L_a(y)u = P\operatorname{diag}(f_1^a(y),\dots,f_q^a(y))u$, where f_i^a are the Gaussian densities. Integration preserves total mass. The future categorical audit separates the coordinates. Pullback of every future effect is another linear functional on the probability cone, so the finite affine test closure holds. The normalized output lies in $K = P\Delta_q \subset \{u_i \ge a_0\}$, which has full dimension by invertibility. There is a positive Euclidean buffer from K to the boundary of D, on the affine probability hyperplane. Use an orthonormal affine chart for the task, so that its quadratic loss is the usual Brier loss.

Write $v_i = u_i f_i^a(y)/\sum_j u_j f_j^a(y)$ for the posterior before P. Its log odds are

$$\begin{aligned} z_i &= \log(v_i/v_q) \\ &= \log(u_i/u_q) + (m_i^a - m_q^a)^T\Sigma_a^{-1}y - \tfrac{1}{2}[(m_i^a)^T\Sigma_a^{-1}m_i^a - (m_q^a)^T\Sigma_a^{-1}m_q^a]. \end{aligned}$$

Under the actual incoming mixture, z has a mixture of nonsingular Gaussian densities. Since $u \in D$ and the action set is finite, their means and covariance bounds are uniform (the constants include a positive lower bound for the absolute determinant of the affine likelihood-rank matrix); thus their density is at most $C\exp(-c\|z\|^2)$. The inverse logistic Jacobian contributes $(\prod_i v_i)^{-1} \le C\exp(C\|z\|)$. Consequently the actual density of v is bounded on the entire simplex, including approaches to its boundary. The invertible affine map P gives a uniform density bound on K. This is an actual-law calculation, not a dimension count.

The report-mixture density is affine in u, and its L^1 difference is at most $\sum_i |u_i - u'_i|$, giving report continuity. Off a common null set the likelihood is positive. The normalized maps are continuous in u, and domination by the sum of the Gaussian densities gives weak continuity of the acquired rows. The common grid and null boundaries follow from the density bound. The categorical audit gives $g(u) = 1 - \|u\|^2$ before the harmless affine chart. All hypotheses have been verified directly from the raw kernel. For the quotient condition, finite action words and compact report boxes have compact attained images: all finite-word updates are continuous and their likelihoods are positive. The union over word lengths, action words and integer report boxes is therefore Borel. On each compact domain its continuous fiber map has a Borel section, obtained by successive selection of intersecting finite closed-ball covers of vanishing diameter; compactness preserves a point in the nested fiber. Choose the first compact image containing a given state to obtain a section on the countable union. No section is used by the implemented observer. Projective filter stability, though available for positive P, is not used. □

Proposition 5.2 (A genuine Gaussian stopping region)**.** *For binary states, transition probability* $1/4$ *to switch, reports* $N(-1,1)$ *and* $N(1,1)$*, Bernoulli squared audit and* $\kappa = 1/100$*, the optimal full-history policy has both a nonempty open continuation region and a nonempty open stopping region inside the acquired interval* $K = [1/4, 3/4]$*. Both have positive probability under an actual first report from any interior preparation.*

Proof. Here $g(u) = u(1-u)$ and $\min_K g = 3/16$. Continuation must incur at least $\kappa + 3/16$, so points satisfying $g(u) < \kappa + 3/16$ are strict stopping points, including neighborhoods inside K of both endpoints.

At $u = 1/2$, let v be the posterior before transition. Conditioning only on the sign of the report gives posterior means $\Phi(1)$ and $1 - \Phi(1)$ with equal probabilities. Hence

$$\mathrm{Var}(v) \geq (\Phi(1) - 1/2)^2 \geq \varphi(1)^2 = \frac{1}{2\pi e}.$$

The second inequality uses $\int_0^1 \varphi(x)\,dx \geq \varphi(1)$. After transition $u' = 1/4 + v/2$, so one call followed by stopping reduces expected audit loss at the center by at least $1/(8\pi e) > 1/100$. Thus continuation is strictly better there. The Bellman alternatives are continuous, so the strict inequality holds on a neighborhood. The Gaussian log-odds law has a strictly positive density throughout the interior of K, proving actual positive mass of both regions. This is not a stopping option that is formally present but never used. □

5.2. **Noncommuting quantum instruments.** Let the physical Hilbert-space dimension be $h \geq 2$, and let H range over the Hilbert–Schmidt unit ball in traceless Hermitian matrices, with normalized volume measure λ_0. Fix $0 < \eta, \lambda < 1$ and a finite family of unitaries U_a. Put

$$E_H = I + \eta H, \quad K_H = E_H^{1/2}, \quad \tau_* = I/h, \tag{20}$$
$$L_a(H)\rho = \lambda U_a K_H \rho K_H U_a^* + (1-\lambda)\,\mathrm{tr}(E_H\rho)\tau_*.$$

The report is H with likelihood $\mathrm{tr}(E_H\rho)$. The effects need not commute. At commitment choose one of a finite informationally complete collection of bounded traceless Hermitian coordinates independently, measure its binary effect and return its rescaled coordinate vector. This is one executable randomized audit with affine second moment and mean equal to the coordinate vector of ρ.

Theorem 5.3 (Quantum optimal acquisition and stopping)**.** *The instruments* (20)*, with the stated audit, satisfy Theorem 3.1 in dimension* $d = h^2 - 1$*. All finite-state control is classical and fully counted. No common eigenbasis, quantum memory for the observer or uniform feedback-overlap estimate is assumed.*

Proof. Each map is completely positive, and $\int E_H\, d\lambda_0 = I$ by symmetry, so integration preserves trace. The linear space of Hermitian effects is finite and invariant under every dual instrument; informationally complete tests separate the density operators. Continuous finite-word maps on the compact report ball have compact images. Their countable union is Borel and has a Borel history section by the compact-fiber construction in the Gaussian proof; append the visible interface status separately. The chosen affine coordinates therefore give the required predictive quotient from the raw experiment. The likelihood lies in $[1-\eta, 1+\eta]$. Normalized outputs belong to

$$K = \{\rho : \rho \geq (1-\lambda)\tau_*,\ \operatorname{tr}\rho = 1\}, \quad D = \{\rho : \rho \geq (1-\lambda)\tau_*/2,\ \operatorname{tr}\rho = 1\}.$$

These convex full-dimensional sets have a uniform buffer in the trace-one hyperplane. Trace-norm estimates give the report-continuity bound, and all normalized maps are continuous.

We verify actual full-dimensional mass, rather than appealing to matrix dimension. Before the final channel, write $\sigma = E^{1/2}\rho E^{1/2}/\operatorname{tr}(E\rho)$ with $\operatorname{tr} E = h$. For fixed positive ρ its inverse is

$$B_{\rho,\sigma} = \rho^{-1/2}(\rho^{1/2}\sigma\rho^{1/2})^{1/2}\rho^{-1/2}, \qquad E = \frac{hB_{\rho,\sigma}^2}{\operatorname{tr} B_{\rho,\sigma}^2}. \tag{21}$$

Indeed $B_{\rho,\sigma}\rho B_{\rho,\sigma} = \sigma$, and the unique positive congruence solution proves both directions. For $\rho \in D$ and $E = I + \eta H$, all spectra of the arguments in (21) stay in fixed compact subintervals of $(0,\infty)$. The inverse and its derivative are uniformly bounded. The chain rule makes the forward derivative invertible with Jacobian bounded away from zero. Multiplying the bounded input volume density by the bounded actual likelihood gives a uniform posterior density. The final depolarizing unitary channel is invertible affine on the trace-one hyperplane, so the same holds on K. The grid has null row boundaries.

For clarity, choose a Hilbert–Schmidt orthonormal traceless Hermitian basis $T_1, \ldots, T_d$. Then $\|T_j\|_{\mathrm{op}} \leq \|T_j\|_{\mathrm{HS}} = 1$; choose $0 < b \leq 1$, so $\|bT_j\|_{\mathrm{op}} \leq 1$. Choose J uniformly, measure effects $(I \pm bT_J)/2$, and report $Y = (d/b)Ze_J$, $Z \in \{-1, 1\}$. Then $\mathbb{E}Y = (\operatorname{tr} T_j\rho)_j$ and $\mathbb{E}\|Y\|^2 = d^2/b^2$. This supplies exactly (5) without a simultaneous measurement of noncommuting observables. All matrix arguments are finite-dimensional; no assertion about unbounded-operator closure is involved. □

6. Singular prepared sectors without global mixing

6.1. **Boundary-reaching informative updates.** The following class has no fixed interior output carrier, and its same-report derivative exceeds one for some histories. It verifies the boundary alternative of the main theorem rather than borrowing the buffered proof's geometric implications.

Theorem 6.1 (A nonuniform informative binary class)**.** *Let the hidden state be binary and let action j choose $0 < \zeta_j < 1$ from a known finite library. First report $S \in \{-1, 1\}$ with*

$$\mathbb{P}(S = s \mid Z = x) = \tfrac{1}{2}\{1 + s\zeta_j(2x-1)\}.$$

Independently draw and report (A, B) with density 2 on $0 < b < 1$, $0 < a < 1 - b$. After these reports, set the next hidden bit to one with probability $A + BZ$. Use the new bit as the protected Bernoulli audit. This positive primitive experiment satisfies Theorem 3.1 on the entire interval $[0, 1]$, for every fixed $\kappa > 0$. No common interior buffer or uniform strict same-report cone contraction is available. For this class the effective finite-bit conclusion of Corollary 8.5 also holds with direct uniform routing constants.

Proof. The joint raw map is the binary transition with columns $(1-a, a)^T$ and $(1-a-b, a+b)^T$ after the diagonal binary-report likelihood. It is positive and normalized when integrated over the two reports and the triangle. Its affine span is the two-dimensional probability cone; the Bernoulli audit separates the scalar coordinate. Given incoming belief u and report s, put

$$v_s(u) = \frac{u(1 + s\zeta_j)}{1 + s\zeta_j(2u-1)}, \qquad F_j(s, a, b; u) = a + bv_s(u).$$

Given b, s, the variable a is uniform on $(0, 1-b)$, so its transformed row has density at most $1/(1-b)$. The marginal density of b is $2(1-b)$. Averaging over b and s therefore bounds every actual row density by 2, uniformly in $u \in [0,1]$. Likelihoods are positive and affine, report TV is Lipschitz, and normalized outputs are continuous in u. Grids have null boundaries. Finite-word compact exhaustions of the open triangle give Borel attained images and history sections as in Theorem 5.1. These are all hypotheses of the binary boundary alternative.

The closure of the actual acquired support reaches both endpoints, so it is not contained in an interior buffered interval. Moreover

$$\partial_u F_j(s,a,b;u) = \frac{b(1-\zeta_j^2)}{\{1+s\zeta_j(2u-1)\}^2}.$$

For $s=-1$, u close to one and b close to one this exceeds one on an open set of reports. As $(a,b) \to (0,1)$, the positive transition approaches the identity; its projective contraction coefficient tends to one, and the preceding diagonal likelihood map is a projective isometry. Thus there is no strict uniform same-report contraction coefficient. This is a statement about the supremum over reports, not a claim that every averaged stability property fails.

Finally the report likelihood ratio between any two incoming beliefs is at most $H = (1+\zeta_*)/(1-\zeta_*)$, where $\zeta_* = \max_j \zeta_j < 1$, and the displayed derivative is at most the same H. Hence the routing proof of Lemma 8.4 applies directly with $L_F = H$, without its buffered derivation. The rational report formula on a compact triangle has a denominator bounded away from zero. Its boundary has zero Lebesgue mass; certified triangle integration and numerical boundary enclosures supply the effective primitive certificate. Apply Corollary 8.5 with these direct constants. □

6.2. **Immutable prepared sectors.** A singular extension must respect what an optimizing stopping rule may select. Merely giving a one-call mixture of an atom and a continuous law does not force the terminal law to retain the continuous weight: an observer may wait for the atom. The following class fixes that issue at the preparation interface.

A single preparation chooses a sector $J \in \{1,\dots,k\}$ with probabilities $w_j > 0$, and reports it once. The sector is immutable. Conditional on $J=j$, actions and reports form a regular experiment of dimension $d_j \geq 1$ as above, or a known atomic experiment requiring only finitely many labels for its task. Embed sector indicators and their task coordinates in mutually separated affine faces of one finite test space. The audit includes the sector and its own protected task. No observer may discard a sector, restart preparation or condition the task on a favorable sector. The sector mark, if retained, occupies ordinary labels.

Theorem 6.2 (Weighted singular acquired law)**.** *For the finite prepared union just described, let $V_\kappa = \sum_j w_j V_{\kappa,j}$ and let $d_* = \max\{d_j : d_j \geq 1\}$. If a positive-dimensional sector is present, then for all sufficiently large M,*

$$c_{\kappa,w} M^{-2/d_*} \leq \mathcal{R}_{\mathrm{aut}}(\kappa, M) - V_\kappa \leq C_{\kappa,w} M^{-2/d_*}. \tag{22}$$

The same holds for checkpoints. More explicitly, allocating m_j live cells to sector j gives an upper of order $\sum_{d_j \geq 1} w_j m_j^{-2/d_j}$, with total labels at most $1 + k + 2\sum_{d_j \geq 1} m_j + C_{\mathrm{atom}}$. The true terminal sector masses are exactly w_j. No global mixing or common smooth ambient acquired density is required.

Proof. A proper policy has $\mathbb{P}(J = j \text{ at commitment}) = w_j$, because its preparation cannot be abandoned and the sector cannot change. Conditioning on J gives the full-history decomposition. Implement the conditional upper construction of Theorem 3.1 on disjoint label sets. Count one pre-preparation label, the k sector-specific initial labels, all live and halting labels, and the finitely many atomic labels. Fixed proportional allocation proves the upper in (22); no optimal allocation formula is claimed.

For any candidate let its conditional costed excess be $e_j \geq 0$. A global decoder with at most M values yields at most M projected readouts in each sector, even when labels are shared. Conditional projection and Lemma 4.3 imply $e_j \geq a_j(MT_j)^{-2/d_j}$, where $T_j = \mathbb{E}[\tau \mid J = j]$. If global excess $\sum_j w_j e_j \leq L_0$, then $\kappa w_j T_j \leq \kappa + 2L_0$ after increasing the common audit bound if necessary. Thus each positive-dimensional contribution is at least a fixed positive multiple of $w_j M^{-2/d_j}$. If the excess is larger, the lower is immediate with a smaller constant. In particular a highest-dimensional sector supplies the displayed rate. The same proof covers checkpoint encoders. The global acquired law lives on a finite union of faces of different dimensions and possibly atoms, so it is genuinely stratified. Fixed actual weights enter every constant and have not been normalized away. □

One obtains raw realizations by preparing one of finitely many Gaussian experiments of different hidden-state sizes, or one of finite quantum sectors with different matrix dimensions, and using a public one-time sector flag. The resulting dynamics have disjoint invariant sectors and therefore fail any feedback-uniform overlap across the whole preparation space. Their within-sector instruments are those proved in Section 5. Arbitrary dynamic rank changes followed by freely optimized stopping are not inferred from this theorem. The retained stopped-instrument article treats a different, prescribed-hazard rank-changing interface [6].

7. An exact acquired-law frontier under a mean budget

The positive-cost theorem concerns a general controlled experiment, but does not identify every constrained frontier by duality. We now derive such a frontier directly for a different, broad primitive class. It includes singular preparations and requires no smoothness, stability or supplied small-ball dimension.

Fix a probability μ on a compact convex set K of possible task means. Each paid acquisition independently prepares a new hidden target conditional on $U \sim \mu$ and reports U. Its audit has conditional mean U and affine second moment $c_0(U)$. For categorical targets the raw kernel is explicitly

$$K(dx', du \mid x) = u_{x'}\mu(du). \tag{23}$$

The previous hidden target is not available again. Each preparation in (23) is a charged acquisition; it is not a free reset added to a correlated experiment. The present target's protected audit is executable at commitment, so its predictive quotient is precisely U up to any stated task equivalence. Require $\mathbb{E}\tau \leq n$, where $n \geq 1$ is a real mean budget. The final readout receives only its finite halting label.

Theorem 7.1 (Exact terminal-law and total-state frontier)**.** *For the raw class* (23)*, a probability ν is an attainable terminal predictive law under a proper adapted stopping rule with $\mathbb{E}\tau \leq n$ if and only if $\nu \leq n\mu$. Put*

$$\Phi_\mu(n,k) = \min_{\nu \in \mathcal{P}(K),\ \nu \leq n\mu} \left\{ \int g\, d\nu + q_k(\nu) \right\}. \tag{24}$$

Then, for every $n \geq 1$, $k \geq 1$ and $M \geq 2$,

$$\mathcal{R}_{\mathrm{cp}}(n,k) = \Phi_\mu(n,k), \qquad \mathcal{R}_{\mathrm{aut}}(n,M) = \Phi_\mu(n,M-1). \tag{25}$$

The minima are attained. The autonomous upper has exactly one live label and at most $M-1$ halting labels. It uses no clock or hidden history register. The lower covers every proper adapted Borel randomized program, including programs with many live labels.

Proof. Let ν_j be the terminal measure on output label j. Independence of each fresh U_t from the pre-call history gives

$$\sum_j \nu_j(A) = \sum_{t \geq 1} \mathbb{P}(\tau = t, U_t \in A) \leq \sum_{t \geq 1} \mathbb{P}(\tau \geq t)\mu(A) = (\mathbb{E}\tau)\mu(A).$$

This proves necessity without assuming a stationary or memoryless stopping rule. Conversely write $h = d\nu/d\mu \le n$. At each call accept u with probability $h(u)/n$, otherwise acquire again. The success probability is $1/n$, so the mean is n and the terminal law is exactly ν. To implement a finite code with submeasures ν_j, accept and halt in j with probability $a_j(u) = d\nu_j/(n\,d\mu)$; their sum is at most one. This one-live-label implementation realizes all the desired submeasures and readouts.

Conditioning the squared audit on U gives $g(U) + \|U - r_j\|^2$. For fixed terminal law, the best code is its usual k-point quantizer. This proves the checkpoint equality and the autonomous upper. Any proper autonomous program has at least one reachable live label, and therefore at most $M-1$ halting labels. Its terminal submeasures satisfy the same domination, proving the autonomous lower.

For attainment it is convenient not to assume continuity of a selected quantizer. Work with $0 \le a_j \le 1$, $\sum_j a_j \le 1$ in the unit ball of $L^\infty(\mu)$, with $\sum_j \int a_j\,d\mu = 1/n$, and readouts in the compact task range. This is weak-star compact times a compact finite-dimensional set. The objective $n\sum_j \int a_j(u)[c_0(u) - 2\langle u, r_j\rangle + \|r_j\|^2]d\mu$ is jointly continuous: varying readouts changes its integrand uniformly, and fixed integrands are in L^1. A minimizer exists. Borel versions of the finitely many densities implement it. Null sets, atoms and singular continuous μ cause no problem. □

Theorem 7.1 is an intrinsic attained characterization in its raw independent-observation class: no global cover or dimension is assumed. Independence in (23) is essential. It is not used to prove Theorem 3.1, and the frontier is not asserted for Gaussian filtering or a general correlated quantum trajectory.

7.1. **A closed joint acquisition–memory curve.** Let U be uniform on $[1/4, 3/4]$ and, conditional on U, let the current hidden target be Bernoulli(U). Use scalar squared loss. Then $g(u) = u(1-u)$ and the primitive report is continuous and informative.

Proposition 7.2 (Sharp mean-acquisition law and exact small budgets)**.** *For $n \ge 1$ the full-history mean-budget Bayes risk is*

$$B_n = \frac{3}{16} + \frac{1}{16n} - \frac{1}{48n^2}. \tag{26}$$

Uniformly in $n \ge 1$ and $k \ge 2$,

$$c(nk)^{-2} \le \Phi_\mu(n,k) - B_n \le C(nk)^{-2}. \tag{27}$$

In particular (27) *holds with $k = M-1$ for autonomous machines with $M \ge 3$, and with k terminal labels for checkpoints. Moreover,*

$$\mathcal{R}_{\mathrm{aut}}(n,2) = \frac{3}{16} + \frac{1}{8n} - \frac{1}{16n^2}, \tag{28}$$

$$\mathcal{R}_{\mathrm{aut}}(n,3) = \frac{3}{16} + \frac{1}{16n} - \frac{1}{64n^2}. \tag{29}$$

These are optima over all randomized adapted stopping programs, not just threshold algorithms.

Proof. Theorem 7.1 bounds the terminal density by $2n$. To minimize the full-history loss g, put maximal density on its lowest values: the two endpoint intervals of equal width $w = 1/(4n)$. This conclusion follows directly by swapping equal amounts of mass from any higher-loss point to an unfilled lower-loss point. Integrating $g(1/4 + x) = 3/16 + x/2 - x^2$ for $0 \le x \le w$ yields (26). It is attained by rejecting all other reports, with acceptance probability $1/n$.

Every terminal law has density at most $2n$. The union-of-balls lower in dimension one gives $q_k \ge c(nk)^{-2}$, while its unquantized audit risk is at least B_n. For the upper retain the same two endpoint intervals and use $\lfloor k/2 \rfloor$ centers on one and the remaining centers on the other. Their cell

lengths are at most $C/(nk)$, proving (27). The one-live-label machine implements both acceptance and the halting code; the expected call count is exactly n.

For one halting readout r, the conditional risk $u(1-2r)+r^2$ is affine in u. Its smallest accepted set is one endpoint interval of length $L = 1/(2n)$, unless the slope vanishes. Its optimal readout is the interval mean $1/4 + L/2$ (or its reflection). The resulting loss is $3/16 + L/4 - L^2/4$, proving (28); the zero-slope constant $1/4$ is no smaller.

For two readouts the lower envelope of their affine risks is concave. Mass swapping therefore makes an optimal accepted set one endpoint interval or a union of two endpoint intervals of total length L. If both tails are present and $L < 1/2$, their rejected gap contains the change between the two active affine pieces. Each tail is served by a single readout, which must be its centroid. Writing their lengths as $x, L-x$, the risk is

$$\frac{3}{16} + \frac{2n}{4}\{x^2 + (L-x)^2 - x^3 - (L-x)^3\}.$$

After $x = L/2 + t$, the coefficient of t^2 in braces is $2 - 3L > 0$. The symmetric choice minimizes it and gives $3/16 + L/8 - L^2/16$, which is (29).

If only one endpoint interval is accepted, even the best two-center code on that uniform interval has risk at least $3/16 + L/4 - 5L^2/16$. Indeed its full-history Bayes integral is $3/16 + L/4 - L^2/3$, and its optimal two-center squared distortion is $L^2/48$, obtained by equal halves; the latter also follows by minimizing the sum of cubed cell lengths. Its difference from the symmetric two-tail value is $L(1-2L)/8 \geq 0$. When $L = 1/2$ there is no rejected gap and the ordinary uniform two-center calculation gives the same value. Ties and randomized acceptance do not improve the linear mass allocation or the pointwise best readout. This exhausts the possible envelopes and proves the all-program optimum. □

7.2. **A singular, nonambient acquisition exponent.** Let $0 < r < 1/2$ and let C_r have the balanced self-similar law generated by $x \mapsto rx$ and $x \mapsto 1 - r + rx$. Prepare $U = 1/4 + C_r/2$, report it on each independent paid call and audit Bernoulli(U). Put $s_r = \log 2/\log(1/r)$.

Corollary 7.3 (Singular acquired frontier)**.** *For $n = 2^{h-1}$, $h \geq 1$, and $k \geq 2$, define $\ell_h = r^h/2$. The full-history mean-budget Bayes value and the finite-code excess satisfy*

$$B_{n,r} = \frac{3}{16} + \frac{\ell_h}{4} - \frac{\ell_h^2}{2(1+r)}, \qquad \Phi_{\mu_r}(n,k) - B_{n,r} \asymp_r (nk)^{-2/s_r}. \tag{30}$$

The same equality and bounds apply to an autonomous observer with $M = k+1$ total states by Theorem 7.1. Its real terminal support is singular in the scalar ambient interval. Constants can be uniform for r in a compact subinterval of $(0, 1/2)$.

Proof. Every depth-j cylinder has length r^j and mass 2^{-j}. Their separation gives the small-ball upper $\mu_r(B(x,t)) \leq C_r t^{s_r}$: choose the cylinder scale comparable to t; an interval of that length meets at most a bounded number of cylinders, with a bound uniform on compact r-ranges. Off-support centers do not alter the bound. Since every stopped law is bounded by $n\mu_r$, a union of k radius-t balls has mass at most $C_r nkt^{s_r}$. This gives $q_k \geq c_r(nk)^{-2/s_r}$ for every allowed stopping program.

The two outermost depth-h cylinders together have exactly $2^{1-h} = 1/n$ of the actual preparation mass. They are exactly the lowest values of the symmetric loss g up to a null boundary. The same mass-swapping argument as in Proposition 7.2 makes them full-history optimal. Conditional on a left cylinder, $U = 1/4 + \ell_h C_r$; the right one is its reflection. The self-similar identities give $\mathbb{E}C_r = 1/2$ and $\mathbb{E}C_r^2 = 1/[2(1+r)]$. Integration gives $B_{n,r}$.

Accept those two cylinders and, inside each, use further binary cylinder cells, with at most k total halting labels. Their diameters are at most $C_r \ell_h k^{-1/s_r}$, so their squared distortion is at most $C_r \ell_h^2 k^{-2/s_r}$. Since $\ell_h = (r/2)n^{-1/s_r}$, this is the asserted upper. The exact single-live-state realization

accepts with probability $1/n$ and has mean call count n. Actual mass was not renormalized without paying for rejected calls. This proves a singular resource law, not merely admissibility of a singular measure in an interface. □

8. Same-task errors, simulation and computational resources

8.1. **An attained joint acquisition–memory–error law.** An error allowance alone is not a risk lower bound. We give a fixed experiment in which the claimed errors are actually present and the protected task is the same on both sides.

Prefix the uniform Bernoulli core of Proposition 7.2 by one paid report of an independent fair bit $B \in \{0,1\}$. This report is erased with probability $a \in [0,1]$, independently of the core. No later report or action reveals B. Add an invisible constant calibration coordinate $\theta \in \{-\delta, \delta\}$, $0 \le \delta \le \delta_0$, with no influence on any observation. The final audit evaluates the sum of squared errors for the core target, B and θ. Optimize worst-case risk over the two calibration signs. For causal experiment comparison, require the virtual exact-bit report to be produced before the core begins, uniformly for both possible bit values; its protected audit remains the actual B.

Corollary 8.1 (Attained four-resource curve)**.** *The prefix source-to-exact-bit causal deficiency, in total variation of the complete report and bit audit, is exactly $\varepsilon = a/2$. Let $N = n+3 \ge 4$ count one preparation call, one bit-report call, at most n expected core calls and one final audit. For $M \ge 10$, the optimum autonomous risk satisfies*

$$\mathcal{R}(N, M, \delta, \varepsilon) - B_{N-3} \asymp \big((N-3)M\big)^{-2} + \delta^2 + \varepsilon. \tag{31}$$

The constants are uniform over the displayed ranges. A checkpoint has the same order with its terminal label budget. The autonomous upper uses at most $4 + 3k$ labels, $k = \lfloor (M-4)/3 \rfloor$, and its mean core call count is exactly n. No state-minimality claim is made for this product construction.

Proof. On an erasure a parameter-independent simulator must output a bit without information about its true value. Uniform guessing has maximal error $a/2$. Averaging the two true-bit experiments forces error at least $a/2$ for any simulator; the event that the virtual report differs from the protected bit audit proves the same total-variation lower. This proves attained deficiency at the required causal deadline.

Conditional on an erasure, B is still fair given all subsequent core observations and allowed actions, including the stopping event, because the entire core is independent of B. Its unavoidable squared loss is $1/4$, giving $a/4 = \varepsilon/2$. Averaging the two invisible calibration signs gives, for any response r, $\frac{1}{2}[(r-\delta)^2 + (r+\delta)^2] = r^2 + \delta^2$; the minimum is exactly δ^2 at zero.

For any joint policy the final core measure is bounded by $n\mu$: the pre-call history may contain the bit report, but each core draw is independent of it and the total mean core calls are at most n. Therefore its core Bayes contribution is at least B_n and its M-value core decoder incurs at least $c(nM)^{-2}$. These lower bounds concern separate coordinates of one additive squared task and hold simultaneously; the calibration average is taken on that same task.

For the upper, use one prefix label and three lanes, for bit reports $0, 1, \perp$. Each lane has one live core label and k halting labels, using the optimal two-tail acceptance and k-cell code of Proposition 7.2. Output respectively $0, 1, 1/2$ on the bit coordinate and zero on calibration. The core risk is at most $B_n + C(nk)^{-2}$. The count is $1 + 3 + 3k = 4 + 3k \le M$, and $k \ge M/6$ for $M \ge 10$. The bit and calibration contributions are exactly $\varepsilon/2$ and δ^2. A checkpoint stores the same terminal lane/code and can do no worse in order. The physical count $N = n + 3$ includes all fixed calls stated above. It is a mean bound, not a pathwise cap. □

8.2. **Cost-aware causal morphisms.** A morphism consists of a parameter-independent causal simulator, its internal state, report transformation, action lift, stopping and clock map, and task readout map. It must preserve which operations are legal. If an unknown parameter family is

being compared, one fixed simulator must work for every parameter; a family of parameter-specific simulators is not a morphism. Target observer coins are part of the simulated complete law.

Proposition 8.2 (Cost-aware risk transfer)**.** *Let a simulator and a target observer induce a source execution and an ideal target execution of the same bounded audit loss, coupled with terminal discrepancy probability at most ϵ and $\mathbb{E}|\tau' - \tau| \le b$. If the terminal loss approximation is at most v in absolute value, then*

$$|J'_\kappa - J_\kappa| \le L_0\epsilon + \kappa b + v. \tag{32}$$

If the simulator uses R persistent states and the target observer uses M, the direct composed state budget is at most RM, together with any separately declared clock states. Program descriptions and workspace add; call and time maps compose. Two-sided risk comparisons require appropriate simulators in both directions and their separate resource maps.

Proof. On coupled identical terminal task data and readouts the audit losses agree. On the complementary event their difference is at most L_0. The acquisition-cost difference is bounded by $\kappa\mathbb{E}|\tau' - \tau|$. Add the declared terminal approximation. Store the simulator and observer states as an ordered pair to implement the lift; their state counts multiply. A one-way construction gives only the corresponding one-way infimum inequality. An unbounded stopping cost cannot be controlled from terminal total variation alone. □

Proposition 8.3 (A finite-precision certificate for the constructed observer)**.** *Fix a nominal observer of Lemma 4.2 with safety floor $s > 0$ and $T = \mathbb{E}\tau$. Suppose a compiled version preserves that stopping floor exactly, and at each coupled cycle consisting of one acquisition and its following live decision the uniform conditional probability of a first discrepancy in the report primitive, cell routing or action/stop coin is at most η. Suppose its readout differs from the nominal calibrated readout in norm by at most v. Then, with a common bound on the perturbed audit loss,*

$$J_\kappa^{\mathrm{comp}} - J_\kappa^{\mathrm{nominal}} \le T\eta(L_0 + 2\kappa/s) + v^2. \tag{33}$$

A uniform primitive/report boundary-mass certificate and certified numerical action probabilities suffice to bound η. The result does not assume a discrete numerical report is TV-close to a continuous raw report.

Proof. Expose the first discrepancy on the nominal actual execution. Summing its conditional probabilities over visits gives probability at most $T\eta$, without conditioning the row-mass estimate on no earlier mismatch. Before it, durations agree. After it, the safety floor bounds each residual mean number of calls by $1/s$, so $\mathbb{E}|\tau' - \tau| \le 2T\eta/s$. Apply Proposition 8.2. On the nominal law, conditional calibration of every halting label makes the cross term between $Y - q_i$ and the fixed readout error vanish. Hence replacing q_i by its approximation adds at most v^2 exactly there. First compare compiled and nominal executions with the same approximated readout, then apply this nominal identity. All resulting law errors are already in the first-mismatch term. □

For a numerical implementation, membership near a grid boundary is controlled by the unconditional actual row mass in its prescribed neighborhood; primitive and arithmetic failures are separately added. Quantizing report values themselves need not be close in total variation. The Gaussian tail, if a bounded numerical workspace is used, is a charged failure event rather than a silently discarded part of the experiment.

8.3. **Finite-bit synthesis without increasing the state count.** The fixed point itself need not be computable from bare Borel data. For effective raw instruments, however, the buffer yields a uniform approximation bound even though the representatives are calibrated only in occupation, not at each deterministic time.

Lemma 8.4 (Buffered report domination and routing continuity)**.** *In the buffered alternative of Definition 2.2, put $K^+ = K + \overline{B}_{r/4}$. There are fixed constants H, L_F such that, off common null sets,*

$$\ell_a(y,u) \le H\ell_a(y,q), \qquad \|F_a(y,q) - F_a(y,q')\| \le L_F\|q - q'\|, \tag{34}$$

for every $u \in D$, $q, q' \in K^+$. One may take $H = 1 + 2\,\mathrm{diam}(D)/r$ and $L_F = 2\,\mathrm{diam}(K)/r$. In particular, if $q \in K$ and $\|q' - q\| \le \rho \le r/4$, the probability under any actual report row from $u \in D$ that the common grid routes $F_a(y,q)$ and $F_a(y,q')$ differently is at most

$$C\Lambda H L_F m^{1/d}\rho. \tag{35}$$

The constants do not require $L_F < 1$ or stability of iterated updates.

Proof. Every $q \in K^+$ has its ball of radius $r/2$ in D. Express $q = tu + (1-t)v$ with $t = (r/2)/(r/2 + \mathrm{diam}\, D)$ and $v \in D$. Positivity and affinity give $\ell(q) \ge t\ell(u)$, proving domination. For any unit direction e, write $q_\pm = q \pm (r/2)e$. Their likelihoods sum to $2\ell(q)$, and their normalized outputs belong to K. Differentiating the affine numerator and denominator gives

$$r\ell(q)\,\partial_e F(q) = \ell(q_+)\{F(q_+) - F(q)\} - \ell(q_-)\{F(q_-) - F(q)\}.$$

The diameter of K bounds each difference. Integrate the derivative along the segment in the convex K^+ to obtain (34).

Routing can change only within $L_F\rho$ of a grid hyperplane. The union of these slabs in the fixed bounding box has volume at most $Cm^{1/d}L_F\rho$. Under the representative's actual row use its density bound; under an arbitrary current-history row use the likelihood domination by H. This last step is necessary: occupation calibration alone would not identify the report law at a particular deterministic visit. It proves the uniform bound (35) used in first-discrepancy coupling. □

Corollary 8.5 (Effective finite-bit upper)**.** *Assume additionally either the buffered routing certificate of Lemma 8.4 or direct uniform report-domination and routing bounds of its form. Assume computable positive bounds for the fixed geometry and task, a computable rational net of K with points in the certified routing domain, and effective integration of the raw instruments over the finitely described routing events, with certified errors in the physical operator norm. Assume report evaluation admits a certified uniform discrepancy tolerance with finite program, workspace and time; these are separately charged. For computable $\kappa > 0$, Theorem 3.1's upper is attained to the same order by a finite-bit program with the same $1 + 2m$ labels. Its observer-table description has length*

$$O_{\kappa,\mathfrak{E}}\big(m(d + |A| + 1)\log(m+2)\big) \tag{36}$$

plus the primitive program and requested report-evaluation certificate. A finite exhaustive synthesis with certified value intervals terminates. No efficiency or matching description/time/workspace lower bound is asserted.

Proof. Use the nominal construction for a sufficiently fine budget and choose dyadic safety s comparable to $m^{-2/d}$, with its earlier constant-factor upper restriction. Let $h = m^{-1/d}$ and $T_* = 2(1 + L_0/\kappa)$. Round each q_i to a net point within ρ, and its action/STOP simplex row to dyadic probabilities with total variation at most C_A/Q, keeping the safety coin separate and exact. Initial sensing actions come from the original finite set and need no rounding. Use the rounded representative as the halting readout. Lemma 8.4 and the report-evaluation certificate give

$$\eta \le C_1 m^{1/d}\rho + C_A/Q + \eta_{\mathrm{prim}}.$$

Choose these quantities so that

$$T_*\eta(L_0 + 2\kappa/s) \le h^2, \qquad \rho^2 \le h^2.$$

For example $\rho \le c_{\kappa,\mathfrak{E}} m^{-5/d}$, $Q \ge C_{\kappa,\mathfrak{E}} m^{4/d}$, and $\eta_{\mathrm{prim}} \le c_{\kappa,\mathfrak{E}} m^{-4/d}$ suffice, with ρ within the declared routing domain (in the buffered case $\rho \le r/4$). Proposition 8.3 bounds the extra costed

risk by $2h^2$. The coarse-budget one-call construction is approximated directly. The dyadic simplex and coordinate encodings give (36); the finite label set is unchanged.

One need not compute the exact fixed point to find such a program. Enumerate the finite representative net, all simplex rows with denominator Q, all initial actions, and the same halting readouts. This finite family contains the rounded construction just proved. Every candidate preserves safety s, so its continuing physical ensemble operator has norm at most $1 - s$ in the cone base norm. Starting from its actual preparation, truncate the occupation sum after H calls; the residual cost is bounded by

$$(1 - s)^H (L_0 + \kappa/s).$$

Effective integration of the finitely many raw subinstruments then encloses the finite sum and terminal linear loss to any prescribed tolerance. Thus every candidate has a certified value interval of width at most h^2. Choose a candidate with smallest certified upper endpoint. Its true value is at most that of the rounded construction plus h^2, proving the claimed order. The search may be exponentially large or worse as a function of the input description; termination rather than efficient complexity is the conclusion. □

For computable Gaussian and quantum parameters bounded away from the declared degeneracies, the effective hypotheses have direct certificates. In the Gaussian case, Gaussian tail bounds first enclose a report box with any requested missed mass; on that box all raw functions, likelihoods and cell inequalities have computable uniform moduli. Positive likelihood has a computable lower bound on the compact box. Box integration and a shrinking slab enclosure of cell boundaries converge, with the boundary mass controlled by Lemma 8.4. The report outside the box is a charged discrepancy, not a discarded part of the physical law. In the quantum case the report domain is compact and the effect and state eigenvalue floors bound the matrix-square-root and normalization derivatives. Finite box integration, including enclosures of the Hilbert–Schmidt ball boundary, gives the required operator intervals. Since raw positive maps are bounded by their mass on a fixed interior cone basis, these mass/error bounds also control the finite-dimensional physical operator norm.

In the explicit uniform example rational n gives rational acceptance thresholds. For computable nonrational n, enlarge acceptance slightly with certified errors to preserve the mean cap and charge the resulting risk change. The Cantor example uses a finite cylinder description at the stated dyadic acquisition budgets; for computable r its numerical tolerances can likewise be charged. Exact Borel formulae for arbitrary real input constants are not thereby finite-bit programs. Workspace stores one current report approximation, one temporary coordinate vector and arithmetic, not a persistent history. A literal grid scan uses $O(md)$ comparisons per report, while rectangular indexing can reduce that cost when certified. Total expected physical time is the per-call evaluation cost times the expected number of calls, plus the separately charged preparation and audit. The offline exhaustive synthesis time is a distinct resource. These statements do not identify persistent labels with total bit complexity.

9. Position of the results and retained dependencies

The general theorem starts from a prepared instrument and a separating family of executable tests. Its finite affine closure, actual-row density, global cells and either a buffer or the binary boundary condition are substantive restrictions. It does not classify every causal experiment or derive finite-dimensional predictive geometry from positivity alone. Within this class it removes a different substantive restriction: the optimal experiment may now choose when to stop, and the proof needs neither feedback-uniform mixing nor a contracting posterior recursion. Positive acquisition cost and actual occupation mass replace that use of stability. The buffered subcase can still imply cone contraction; the boundary alternative and its raw informative realization remove that particular implication. Neither case claims that all forms of averaged stability fail. The

independent prepared-observation theorem is stronger in a different direction: it characterizes the entire acquired terminal-law set and the exact total-state frontier, even for singular μ, but assumes independent charged preparations.

The comparison with prior work should not rest on terminology. Value-directed belief compression studies how posterior approximations affect future decision value [5]. Quantized partially observed stopping already gives computable value approximations and error estimates [7]. Finite belief-MDP approximation establishes nearly optimal policies under weak continuity in discounted problems [8]. Joint quantizer and sequential-detector design is itself an established active-control problem, with nonconvex synthesis and performance bounds [3]. Classical sequential vector quantization also optimizes recursively generated codes [1]. None of these themes is claimed as newly introduced here.

Bayes-risk quantization and its Bregman interpretation are likewise prior ideas [11]; the projection identity is not the novelty claimed here. Kim's recent finite-history analysis is explicitly a fixed-policy, same-execution Wasserstein comparison [9]. Our benchmark instead optimizes acquisition and stopping, and our lower constrains every finite decoder at the actual stopping time. Partition-observation filtering with atomic and continuous components has also been developed in a measure-first formulation [10]; the explicit hard-total-state lower and first-run occupation calibration distinguish the present claim, not the use of a normalized posterior or a mixed measure.

Our principal claim is the specific matched hard-total-state theorem with endogenous stopping, proved by calibrated killed occupation and cost absorption, and its all-stopping acquired-mass converse. It is not merely convergence of an approximate full belief-MDP on an uncharged real register. The exact independent-preparation frontier additionally determines what every adapted stopping program can achieve before one implements it by a single live state. Standard rejection sampling and conditional projection enter that proof; the joint constrained value and state-count equality are what the argument establishes here.

Occupation balance must also be distinguished from actual execution. The literature on absorbing MDPs explicitly studies uniform absorption, compactness and possible failures of measure characterizations [2]. Our auxiliary positive safety probability makes the finite ensemble resolvent unambiguous. We never remove that probability inside a balance equation and declare the resulting invariant mass to be a first-run occupation. The cost bound is obtained before choosing a sufficiently small positive probability.

The complete retained article [6] proves a global routing/disintegration criterion for prescribed controlled report-dependent hazards, preparation and terminal separation, and matching overlap-based bounds; its source and full mathematical text are preserved. Its preceding geometric-stop, finite-time, likelihood-curvature, simulator and realization articles remain complete companions. The mass–curvature argument and preparation-injected calibration mechanism are inherited and reproduced where needed, not relabeled as wholly new inventions. The original fixed-time clock obstruction is not extended to free stopping: the one-live-state constructions here demonstrate why such an extension would be false.

Unknown-kernel learning and regret, arbitrary nondominated or infinite-dimensional predictive carriers, nonsummable sensitivity lower curves without a coercive cost, arbitrary dynamically selected singular strata, and a matched joint lower frontier for program length, workspace and physical time remain separate mathematical questions. Neither the two raw realizations nor the exact independent-preparation example supplies a converse classification for them. All finite checks accompanying this article are arithmetic and delivery controls, not substitutes for the continuous-parameter proofs above.

Appendix A. Why occupation and interface hypotheses matter

Proposition A.1 (Failure of a cost-free uniform memory gap)**.** *In the uniform Bernoulli experiment of Proposition 7.2, with zero acquisition cost and no mean cap, the infimum risk of a three-state autonomous machine is* $3/16$*, equal to the full-history infimum. Nevertheless each fixed stopped terminal law with bounded density has positive two-point quantization distortion.*

Proof. Accept the two endpoint intervals of width $h > 0$ and use their centroids as the two halting readouts. One live label suffices; $\mathbb{E}\tau = 1/(4h)$. The risk is $g(1/4 + h/2) = 3/16 + h/4 - h^2/4$, tending to $3/16$. Every terminal unquantized risk is at least $3/16$. Its attained terminal density is $1/(2h)$, which diverges as $h \downarrow 0$. Thus neither a one-call density bound without occupation mass nor constants uniform at zero acquisition price can prove a positive excess gap. No proper finite-mean procedure attaining the limiting edge law has been asserted. □

Example A.2 (No common overlap is needed)**.** Take two actions in an independent Bernoulli prepared experiment. Action zero draws U uniformly on $[1/5, 2/5]$ and action one uniformly on $[3/5, 4/5]$, then prepares Bernoulli(U) and reports U. The raw kernels are $K_a(dx', du \mid x) = u_{x'}\mu_a(du)$; they are normalized and positive. On the scalar task base choose $K = [1/5, 4/5]$ and a larger compact interior interval $D \subset (0, 1)$. The row density, buffer and finite affine conditions of Theorem 3.1 hold. Yet policies always selecting different actions have disjoint predictive terminal supports after any fixed number of calls. No positive common feedback-overlap measure exists for all such policies. This separates a removed hypothesis from the theorem, rather than serving as its only realization.

Example A.3 (Singular mass can be selected away)**.** If independent paid observations have preparation $\mu = w\mu_{\mathrm{cont}} + (1 - w)\delta_{u_*}$, $w < 1$, an observer can accept only the atom and stop in one output label, with mean $1/(1 - w)$. Thus a positive continuous one-call component does not impose a continuous terminal component under arbitrary optimized stopping. The immutable-sector condition in Theorem 6.2 and the exact domination condition in Theorem 7.1 account for this selection in different, explicit ways. An ambient dimension or one-step mixture weight alone does not.

Example A.4 (Total variation does not price unbounded duration)**.** Two protocols may have identical terminal reports and audit while one adds an independent waiting time with arbitrarily large finite mean. Their terminal total variation is zero, but their costed risks differ by κ times that mean. Even total variation on complete trajectories alone does not bound expectation of an unbounded clock without an integrability certificate. This explains the separate duration term in Proposition 8.2. It also prevents a fixed-time simulator estimate from being silently promoted to an unbounded stopping theorem.

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